



The Science of Data: Understanding Statistics

Statistics is the branch of mathematics that deals with **collecting, organizing, analyzing, interpreting, and presenting data** to draw meaningful conclusions and support decision-making.

In simple words: It is the **science of learning from data**.

The Dual Pillars of Statistics

Statistics has two main parts, each serving a distinct purpose in understanding and utilizing data.

Descriptive Statistics

Deals with **summarizing and describing** the main features of a dataset.

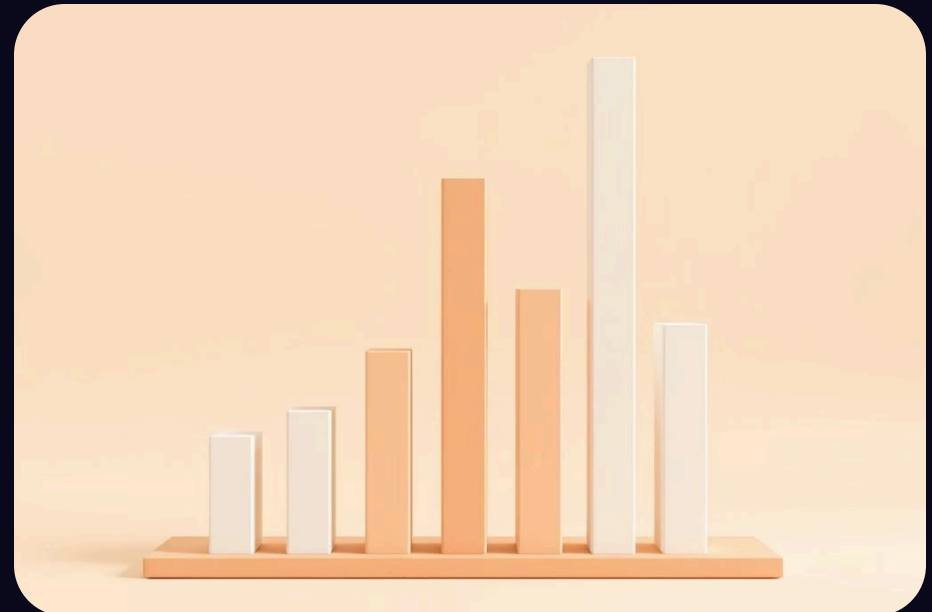
Example: mean, median, mode, standard deviation, graphs, charts.



Inferential Statistics

Deals with **making predictions or generalizations** about a population based on a sample.

Example: hypothesis testing, confidence intervals, regression, t-tests, ANOVA.



Fundamental Concepts in Statistical Analysis

Before diving deeper, it's crucial to grasp these core terms that form the foundation of statistical thinking.



Population

The **entire group** of individuals, objects, or measurements we are interested in.

Example: All students in your college.



Sample

A **subset of the population** that is actually studied.

Example: 50 students selected randomly from your college.



Parameter

A **numerical value that describes a characteristic of the population.**

Usually unknown, fixed.

Example: The true average height of all students in your college.



Statistic

A **numerical value calculated from a sample.**

Used to estimate the population parameter.

Example: The average height of the 50 sampled students.



Descriptive Statistics: Your Starting Point

Descriptive statistics helps us to **summarize raw data** into a meaningful form. It mainly includes:



Measures of Central Tendency

(tell us about the center)

- **Mean** → average
- **Median** → middle value
- **Mode** → most frequent value



Measures of Dispersion (Variability)

(tell us about the spread)

- **Range** → difference between highest & lowest
- **Variance** → average squared deviation
- **Standard Deviation (SD)** → spread around mean



Measures of Position

- **Percentiles** (e.g., 90th percentile = score below which 90% fall)
- **Quartiles** (Q1, Q2, Q3 dividing data into 4 parts)

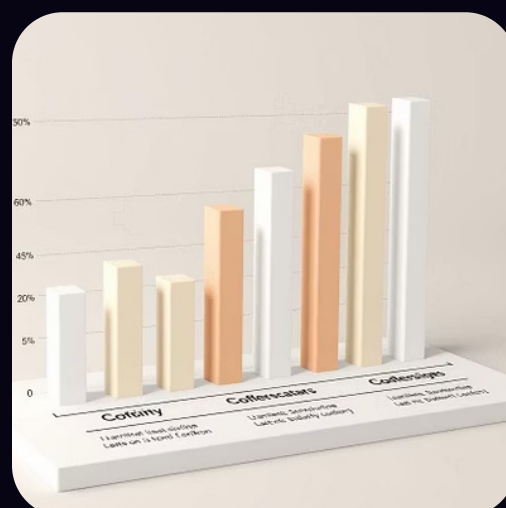
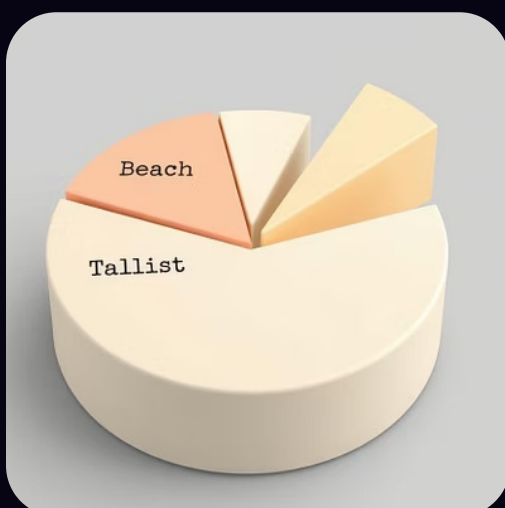


Data Presentation

Visualizing data for clarity and impact.

- **Tables**
- **Graphs**
- **Charts** (Histogram, Pie chart, Bar chart, Box plot)

Effective data presentation transforms raw numbers into compelling insights.



Measures of Central Tendency

These statistical measures identify the central or typical value of a dataset, providing a single number that summarizes the entire distribution.



Mean (Arithmetic Average)

The sum of all values divided by the total count. It uses every data point but is sensitive to extreme values.



Median (Middle Value)

The central value when data is arranged in order. It's robust to outliers and ideal for skewed distributions.



Mode (Most Frequent)

The value that appears most often in a dataset. Useful for categorical data or identifying common occurrences.



Quick Guide: Use **Mean** for normal data without outliers, **Median** for skewed data, and **Mode** for categorical insights.